

Task Submission Document - Iris Classification

Task No: ____ | Student Name: Jeswanth Reddy Reddem | Date: _____

Important: Before final submission, replace all placeholder links below with your actual GitHub, Google Drive, and notebook links. Ensure Google Drive sharing is set to "Anyone with the link can view".

1-Page Executive Summary

Project Goal: Build a supervised machine learning classification model to predict iris flower species using sepal length, sepal width, petal length, and petal width.

Dataset: Classic Iris dataset containing three species: Iris-setosa, Iris-versicolor, and Iris-virginica. The dataset was split into training and testing sets using stratification to preserve class distribution.

EDA and Visualization: Exploratory data analysis was performed using class distribution charts and feature scatter plots. Petal length and petal width showed strong class separability, especially for Iris-setosa.

Models Trained: k-Nearest Neighbors, Logistic Regression, and Decision Tree classifiers were trained and compared.

Evaluation Metrics: Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The best model was selected based on test accuracy and macro/weighted performance.

Best Model: Logistic Regression was selected as the best model in this run, achieving 93.33% test accuracy. The trained model was saved using both joblib and pickle formats.

Deliverables: The submission contains a well-structured notebook, source code, saved model files, generated plots/screenshots, README, requirements file, and inference example.

Submission Links

Artifact	Required Link	Status
GitHub Repository	Paste GitHub repo link here	To be added
Notebook Link	Paste GitHub/Colab notebook link here	To be added
Saved Model File	Paste Drive/GitHub model file link here	To be added
Screenshots/Plots Folder	Paste Drive/GitHub screenshots folder link here	To be added
Final Project ZIP	Paste Google Drive ZIP link here, if required	Optional

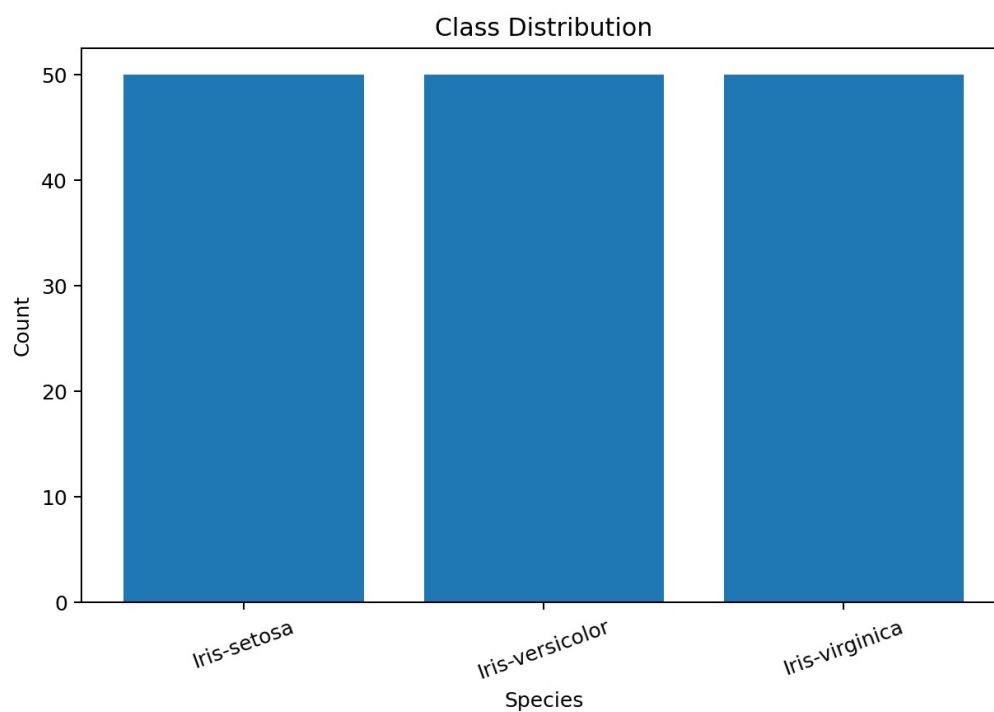
Artifacts Included

- Notebook: notebooks/Iris_Classification.ipynb

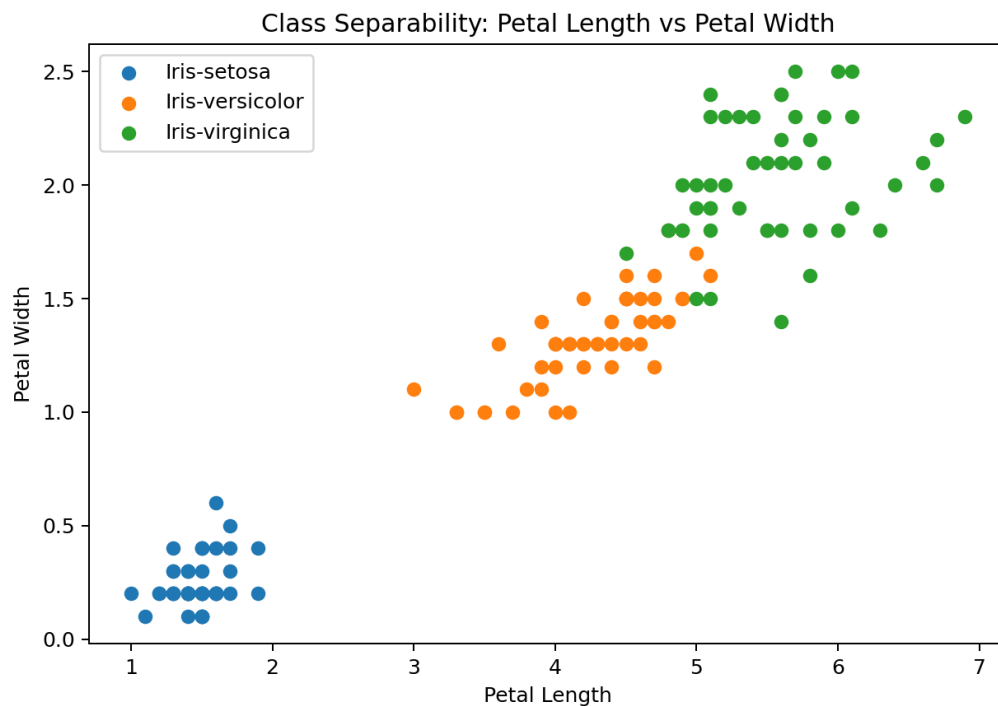
- Training script: train_model.py
- Inference script: inference.py
- Saved models: models/best_iris_model.joblib and models/best_iris_model.pkl
- Plots/screenshots: plots/class_distribution.png, plots/petal_scatter.png, plots/sepal_scatter.png, plots/metrics_comparison.png, plots/confusion_matrix_best_model.png
- README: README.md
- Requirements: requirements.txt
- Metrics summary: metrics_comparison.csv

Screenshots / Plots

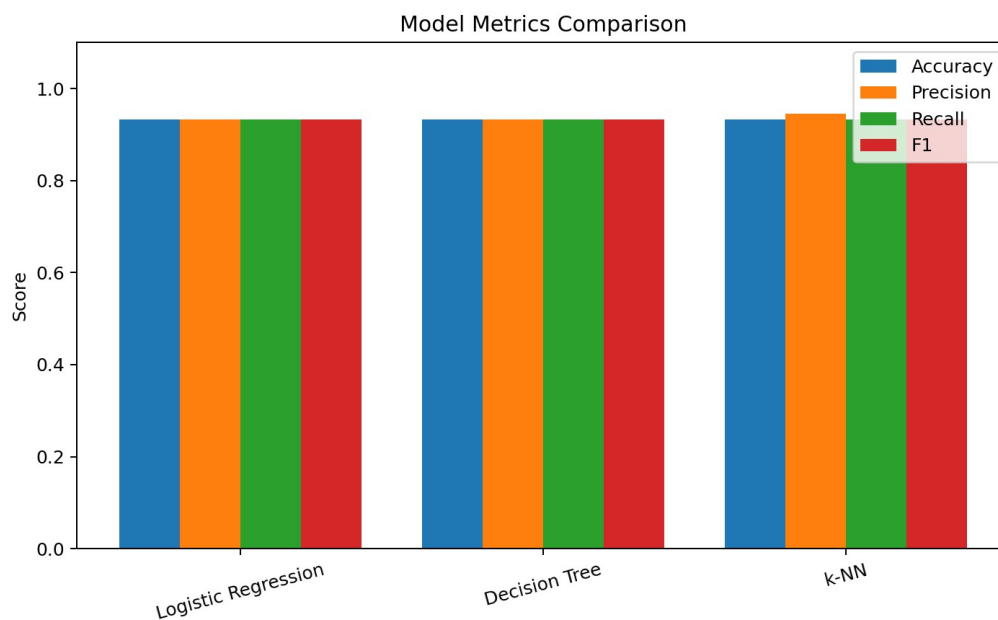
Class Distribution



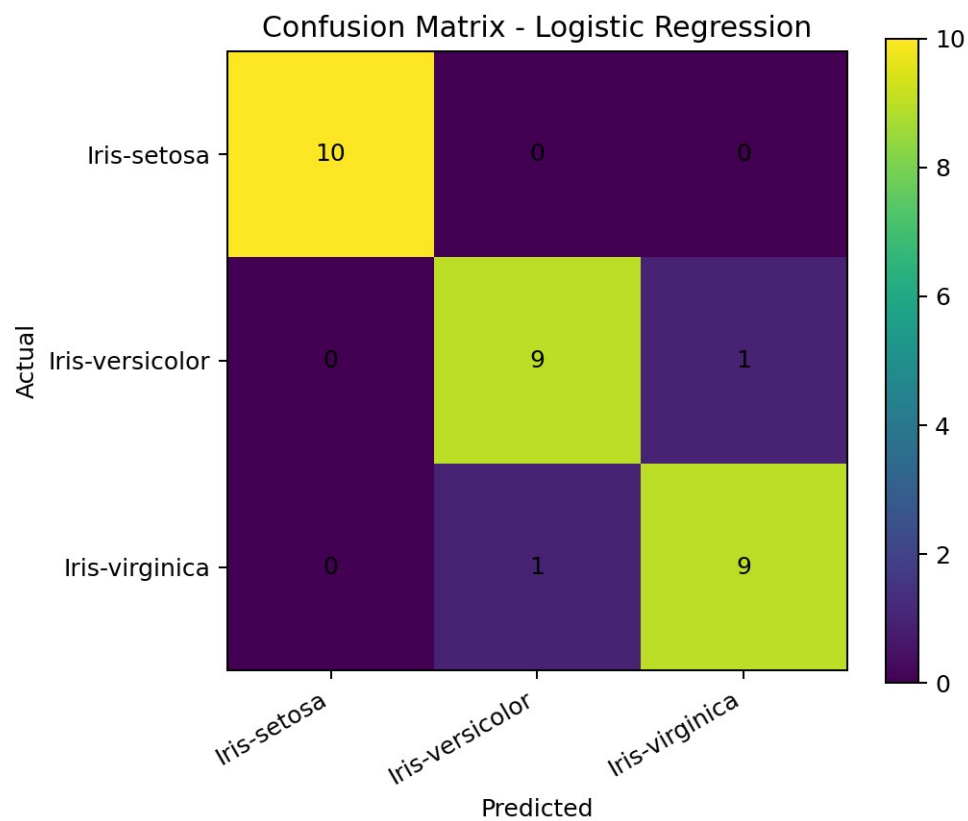
Petal Feature Separability



Metrics Comparison



Confusion Matrix - Best Model



Reproducibility Commands

1. Install dependencies

```
pip install -r requirements.txt
```

2. Train and evaluate models

```
python train_model.py
```

3. Run example inference

```
python inference.py
```

4. Open the notebook

```
jupyter notebook notebooks/Iris_Classification.ipynb
```

Package Versions

- Python 3.x
- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn
- joblib
- jupyter

Final Submission Checklist

- ☐ Push notebooks and code to GitHub.
- ☐ Upload large files or final ZIP to Google Drive if required.
- ☐ Set Google Drive sharing permission to anyone with the link can view.
- ☐ Paste links clearly on the Task Submission page with task number.
- ☐ Attach or upload this DOC/PDF submission document as required.
- ☐ Verify that README includes package versions and exact commands.